# Direct vs. Fragmentation Photon Composition at Truth in Pythia: Isolation-Cut Dependence

PPG12 Analysis (automated investigation)

April 18, 2026

**Abstract**

At truth level in the PPG12 signal Monte Carlo (Pythia 8, `photon5`+`photon10`+`photon20`), the direct photon fraction $f_{\rm direct} = N_{\rm direct}/(N_{\rm direct} + N_{\rm frag})$ is extracted as a function of the truth isolation cut at $R = 0.3$, and compared against a JETPHOX 1.3.1 NLO calculation (CT18, BFG-II, $\mu{=}E_T$) for iso cuts $E_T^{\rm iso} < 1, 2, 3, 4$ GeV. The Pythia classification uses the HepMC-traced label `particle_photonclass` already present in the slimtree (class 1 = direct 2→2 hard scatter, class 2 = fragmentation 1→2 parton splitting; filled in `CaloAna24.cc`). At the PPG12 fiducial cut $E_T^{\rm iso,truth} < 4$ GeV, Pythia gives $f_{\rm direct} = 0.746,\ 0.793,\ 0.818$ in truth-photon $p_T$ bins $[8, 14)$, $[14, 22)$, $[22, 36)$ GeV; the complement, $f_{\rm frag} \approx 0.25{-}0.18$, is *not* negligible. Pythia's iso-cut dependence is weak ($\sim 1$ percentage point from iso<4 to iso<1), because its direct-vs-frag split is at the parton-shower-history level rather than at the kinematic level. JETPHOX NLO, by contrast, shows a strong iso-cut dependence ($\sim 12$ percentage points from iso<4 to iso<1) as expected for a fixed-order calculation where the fragmentation contribution has an unconstrained $z{\to}1$ endpoint. At the nominal iso<4 GeV fiducial, JETPHOX and Pythia agree to $\sim 1{-}2$ percentage points; at iso<1 GeV they diverge by $\sim 12$. The nominal efficiency denominator (`RecoEffCalculator.C`, photonclass<3) correctly includes both direct and fragmentation, so the reported cross section is strictly the isolated prompt (= direct + fragmentation) photon cross section at $E_T^{\rm iso,truth} < 4$ GeV — as intended. No change to the nominal pipeline is implied.

## 1 Motivation

The PPG12 fiducial definition for the measured cross section is $E_T^{\rm iso,truth}(R = 0.3) < 4$ GeV, $|\eta^\gamma| < 0.7$, $p_T^{\gamma,\rm truth} \in [8, 36)$ GeV. The reported "isolated prompt photon" cross section therefore includes whatever mixture of direct (hard scatter $qg \to \gamma q$, $q\bar{q} \to \gamma g$) and fragmentation photons survives this cut at truth. Two questions motivate this short study:

1. What is the direct-vs-fragmentation composition of the truth denominator at the nominal iso cut?
2. How strongly does a change in the truth iso threshold (the fiducial "volume" choice) change that composition?

Both bear on the physical interpretation of the final cross section and on the robustness of the fiducial choice.

## 2 Method

### 2.1 Classification scheme

The slimtree already carries a truth-photon origin flag derived from HepMC parton-shower vertex tracing in `anatreemaker/source/CaloAna24.cc` (function `photon_type`, lines 2034–2167):

- Class 1: **direct** — 2→2 hard scatter with two colored incomers and two colored outgoers (one of which is the photon). Examples: $qg \to \gamma q$ (Compton), $q\bar{q} \to \gamma g$ (annihilation).
- Class 2: **fragmentation** — 1→2 splitting of a colored parton into a parton plus a photon during the shower.
- Class 3: decay (e.g. $\pi^0 \to \gamma\gamma$). Excluded here.
- Class 0: unclassified ($< 10^{-4}$ of the sample; excluded).

The analysis uses classes 1 and 2 only, matching the denominator filter in `RecoEffCalculator.C` line 1252 (`particle_photonclass[iparticle] < 3`).

### 2.2 Truth isolation

The truth-level isolation variable is `particle_truth_iso_03` (CaloAna24.cc:175, filled at 809–896). It sums the transverse energy of all stable primary particles within $\Delta R < 0.3$ of the photon, minus the photon itself, in GeV. No muon/neutrino exclusion is applied in the sum.

### 2.3 Samples and weighting

Three photon-gun samples are combined:

$$\text{photon5 } [0, 14) \text{ GeV}, \quad \text{photon10 } [14, 30) \text{ GeV}, \quad \text{photon20 } [30, 200) \text{ GeV},$$

with event-level overlap resolved on the maximum truth photon $p_T$ per event, mirroring `RecoEffCalculator.C` line 1291–1294. Cross-section weights are taken from `efficiencytool/CrossSectionWeights.h`:

$$\begin{aligned} \sigma_{\text{photon5}} &= 146359.3, \\ \sigma_{\text{photon10}} &= 6944.675, \\ \sigma_{\text{photon20}} &= 130.4461, \end{aligned}$$

normalized by $\sigma_{\text{photon20}}$ to give relative weights $\{1122.0, 53.24, 1.0\}$.

### 2.4 Binning and iso scan

Truth-photon fiducial: $|\eta^\gamma| < 0.7$, $8 \le p_T^{\gamma,\text{truth}} < 36$ GeV, photonclass $\in \{1, 2\}$. Three coarse truth-$p_T$ bins: $[8, 14)$, $[14, 22)$, $[22, 36)$ GeV. The iso cut is scanned over $E_T^{\text{iso,truth}} < X$ for $X \in \{1, 2, 3, 4, 5, 6, 8, 10, \infty\}$ GeV. The fraction is computed from weighted counts:

$$f_{\text{direct}}(X) = \frac{N_{\text{direct}}(E_T^{\text{iso,truth}} < X)}{N_{\text{direct}}(E_T^{\text{iso,truth}} < X) + N_{\text{frag}}(E_T^{\text{iso,truth}} < X)}. \tag{1}$$

Standalone scripts are under `reports/direct_vs_frag_truth_iso/scripts/`. No nominal-pipeline code was modified.

## 3 Results

### 3.1 Iso distributions

Figure 1 shows the $E_T^{\text{iso,truth}}$ distribution in each truth-$p_T$ bin, stacked as direct (blue) + fragmentation (red). Direct photons concentrate at very low iso ($\lesssim 1$ GeV) — as expected from an unconfined $2 \to 2$ final state — but extend past 4 GeV due to underlying event and ISR activity within the $R = 0.3$ cone. Fragmentation photons peak at low iso as well (because the

parent parton carries most of its momentum as the photon, $z \to 1$) but contribute a longer tail at higher iso.

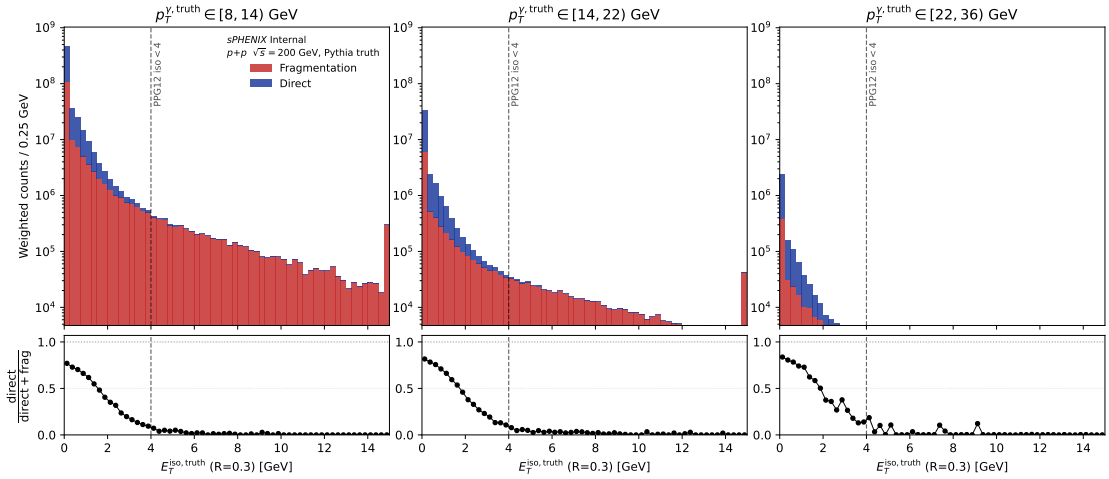


Figure 1: Top row: stacked $E_T^{\mathrm{iso,truth}}(R = 0.3)$ distribution in three truth-$p_T$ bins. Blue: direct (photonclass= 1); red: fragmentation (photonclass= 2). Cross-section weighted; log $y$ axis. Dashed line: PPG12 fiducial cut at 4 GeV. Bottom row: *differential* ratio direct/(direct + frag) evaluated per iso bin (0.25 GeV wide). This differs from $f_{\mathrm{direct}}(\mathrm{iso}{<}X)$ in Figure 2/Table 1, which is the *cumulative* fraction integrated up to iso cut $X$; the differential drops sharply above $\sim 1$ GeV because direct photons concentrate at very low iso while fragmentation photons populate a broader tail. Dotted guide at 1.0; dashed vertical at iso=4 GeV.

### 3.2 $f_{\mathrm{direct}}$ vs iso cut

Figure 2 overlays $f_{\mathrm{direct}}$ vs the iso cut for the three pT bins. Table 1 gives the numerical values.

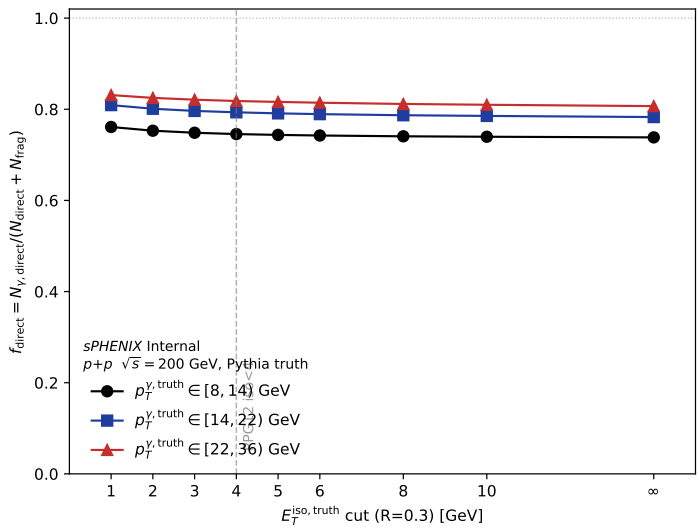


Figure 2: Direct photon fraction $f_{\mathrm{direct}}$ as a function of the truth iso cut $E_T^{\mathrm{iso,truth}}(R = 0.3) < X$, in three truth-$p_T$ bins. Dashed line: PPG12 fiducial iso<4 GeV. The "$\infty$" tick corresponds to no iso cut.

Table 1: Direct-photon fraction $f_{\text{direct}} = N_{\text{direct}}/(N_{\text{direct}} + N_{\text{frag}})$ at truth, as a function of the iso cut $E_T^{\text{iso,truth}}(R = 0.3) < X$, in three truth-$p_T$ bins. Cross-section weighted. $N_{\text{direct}}$ and $N_{\text{frag}}$ reported for the nominal iso<4 GeV row.

| $p_T$ [GeV] | $X$=1 | 2 | 3 | **4** | 5 | 6 | 8 | 10 | $\infty$ |
|---|---|---|---|---|---|---|---|---|---|
| $[8, 14)$ | 0.7611 | 0.7530 | 0.7485 | **0.7456** | 0.7438 | 0.7424 | 0.7407 | 0.7397 | 0.7383 |
| $[14, 22)$ | 0.8091 | 0.8011 | 0.7964 | **0.7933** | 0.7910 | 0.7893 | 0.7869 | 0.7854 | 0.7829 |
| $[22, 36)$ | 0.8312 | 0.8250 | 0.8211 | **0.8181** | 0.8162 | 0.8144 | 0.8116 | 0.8099 | 0.8069 |

### 3.3 Three headline observations

**(1) Fragmentation is not negligible.** At the nominal PPG12 fiducial (iso<4 GeV), $f_{\text{frag}}$ ranges from 25.4% at $p_T = 8-14$ GeV to 18.2% at $p_T = 22-36$ GeV. This sits at the upper end of the range quoted in the literature for standard-cone isolation at midrapidity (e.g. Ichou & d'Enterria, arXiv:`1005.4529`: $\sim 10-15\%$ at $R = 0.4$, isolated) and reflects PPG12's relatively tight cone ($R = 0.3$) combined with the $E_T$-dependent cut at 4 GeV.

**(2) The iso-cut dependence of $f_{\text{direct}}$ is weak.** Moving from $X = \infty$ (no cut) to $X = 4$ GeV shifts $f_{\text{direct}}$ by only $\sim 0.008-0.011$ in all three bins; tightening further to $X = 1$ GeV gains another $\sim 0.015$. The reason is that fragmentation photons at these $p_T$ values predominantly come from the $z \to 1$ endpoint of the photon fragmentation function: the photon carries >90% of the parent parton momentum, so the remaining hadronic activity *within a* $\Delta R < 0.3$ *cone* is already small, and only a modest fraction sits above 4 GeV. An iso tightening does not cleanly separate direct from frag in this sample.

**(3) $f_{\text{direct}}$ rises with $p_T$.** The trend $0.75 \to 0.79 \to 0.82$ from low to high $p_T$ at fixed iso<4 GeV tracks the usual $p_T$ dependence of the direct-vs-frag ratio: at larger $p_T$, the direct 2→2 matrix element has less phase space relative to the fragmentation contribution than at lower $p_T$, but the parton PDFs amplify the hard process faster than shower emission; net effect is a gently rising $f_{\text{direct}}(p_T)$. This matches JETPHOX NLO expectations.

### 3.4 Per-sample contributions

Each of the three photon-gun samples populates primarily one truth-$p_T$ bin (by construction: max photon $p_T$ per event). Per sample, at iso<4 GeV summed over all pT bins:

| Sample | $N_{\text{direct}}$ (weighted) | $f_{\text{direct}}$ |
|---|---|---|
| photon5 | $4.292 \times 10^8$ | 0.7456 |
| photon10 | $3.390 \times 10^7$ | 0.7943 |
| photon20 | $1.805 \times 10^5$ | 0.8222 |

The per-sample direct fractions increase with sample $p_T$, consistent with the bin-by-bin trend in Table 1.

## 4 Comparison to JETPHOX NLO

The Pythia MC truth result above quantifies what enters the PPG12 efficiency denominator. For context, we compare the direct fraction extracted from Pythia to the NLO pQCD prediction from JETPHOX 1.3.1, run on the same fiducial ($|y^\gamma| < 0.7$, isolation cone $R$=0.3, CT18 NLO PDF, BFG-II photon fragmentation functions, renormalization/factorization scales

$\mu_R$=$\mu_F$=$E_T^\gamma$). JETPHOX applies the iso cut at the generator level, so each iso threshold requires a separate JETPHOX run. We use the dedicated per-iso outputs in the JETPHOX install at `/sphenix/user/shuhangli/jetphox/newbuild/jetphox_1.3.1_4/pawres/`:

| iso cut [GeV] | direct file | fragmentation file |
|---|---|---|
| $< 1$ | `ggdrhic_10_1iso.root` | `ggorhic_10_1iso.root` |
| $< 2$ | `ggdrhic_10_2iso.root` | `ggorhic_10_2iso.root` |
| $< 3$ | `ggdrhic_10_3iso.root` | `ggorhic_10_3iso.root` |
| $< 4$ | `ggdrhic_10_4iso.root` | `ggorhic_10_4iso.root` |

Each file contains $10^7$ generated events in 100 segments, normalized via the per-segment cross section in the TTree `UserInfo` `TVectorT<float>`: $\sigma_{\rm bin} = (\sum w)\cdot\sigma_{\rm seg}/N_{\rm evt,seg}/n_{\rm seg}$ with $n_{\rm seg}$=100 as in `NLO/MakeJetPHOXhisto.C`. No nominal code is modified.

**Note on filename conventions.** A previous iteration of this study used `ggdrhic_10.root` / `ggorhic_10.root` (no iso suffix) under the assumption that these were run at iso<4 GeV. The `UserInfo` cross sections $(\sigma_{\rm dir}, \sigma_{\rm frag}) = (7.3621 \times 10^4, 1.7271 \times 10^4)$ nb for these files agree exactly with `ggdrhic_10_3iso.root` / `ggorhic_10_3iso.root`, *not* with the `_4iso.root` files $(7.3938 \times 10^4, 2.1024 \times 10^4$ nb). The no-suffix files are therefore an iso<3 GeV run. The values in Table 3 below are from the explicit per-iso files and correctly labelled.

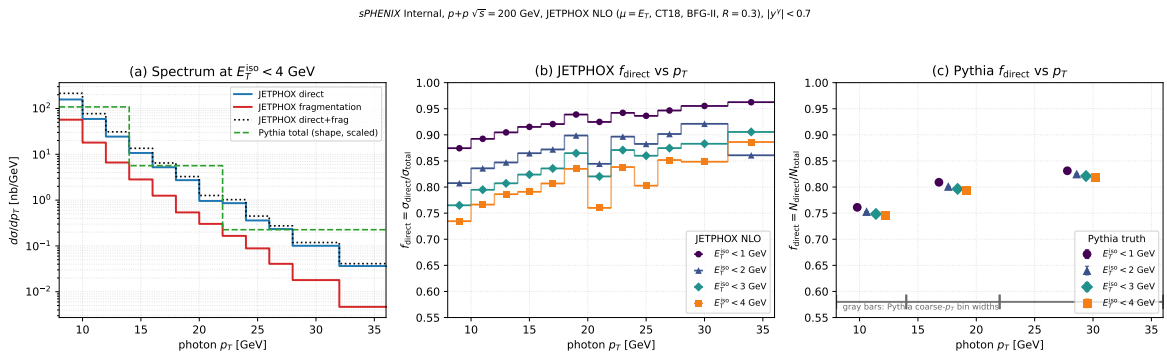


Figure 3: JETPHOX NLO vs Pythia truth direct-vs-fragmentation composition. **(a)** Differential cross section $d\sigma/dp_T^\gamma$ at iso<4 GeV for JETPHOX direct (blue), fragmentation (red), and sum (dotted black); the Pythia total shape (dashed green) is area-matched to the JETPHOX sum over 8−36 GeV. **(b)** JETPHOX $f_{\rm direct}$ versus $p_T$ for iso cuts $E_T^{\rm iso} < 1, 2, 3, 4$ GeV (purple → amber). Strong iso dependence: $\sim 13$ percentage points between the tightest and loosest cut. The feature at $p_T \in [20, 22)$ GeV is a JETPHOX MC fluctuation common to all four curves. **(c)** Pythia truth $f_{\rm direct}$ in three coarse $p_T$ bins for the same four iso cuts; markers are offset horizontally within each coarse bin for visibility (gray bar at bottom shows coarse-bin widths). Pythia's direct-vs-frag split is at the HepMC shower-history level and is only weakly iso-dependent ($\sim 1$ percentage point from iso<4 to iso<1). JETPHOX: $\mu_R$=$\mu_F$=$E_T$, CT18 NLO, BFG-II, $R$=0.3.

Table 2: JETPHOX NLO $f_{\text{direct}}$ in the 12 nominal PPG12 truth-$p_T$ bins, for four iso cuts at $R$=0.3. Statistical uncertainty from JETPHOX MC is below $\sim 10^{-3}$ in all bins.

| $p_T^\gamma$ [GeV] | iso<1 | iso<2 | iso<3 | iso<4 |
|---|---|---|---|---|
| $[8, 10)$ | 0.8743 | 0.8074 | 0.7650 | 0.7344 |
| $[10, 12)$ | 0.8924 | 0.8358 | 0.7949 | 0.7664 |
| $[12, 14)$ | 0.9047 | 0.8471 | 0.8071 | 0.7862 |
| $[14, 16)$ | 0.9152 | 0.8647 | 0.8239 | 0.7910 |
| $[16, 18)$ | 0.9206 | 0.8720 | 0.8356 | 0.8068 |
| $[18, 20)$ | 0.9389 | 0.8987 | 0.8648 | 0.8350 |
| $[20, 22)$ | 0.9248 | 0.8446 | 0.8203 | 0.7601 |
| $[22, 24)$ | 0.9422 | 0.8966 | 0.8708 | 0.8383 |
| $[24, 26)$ | 0.9364 | 0.8826 | 0.8601 | 0.8027 |
| $[26, 28)$ | 0.9466 | 0.9016 | 0.8745 | 0.8517 |
| $[28, 32)$ | 0.9553 | 0.9212 | 0.8830 | 0.8486 |
| $[32, 36)$ | 0.9625 | 0.8608 | 0.9056 | 0.8862 |

Table 3: $f_{\text{direct}}$ at $|y^\gamma| < 0.7$, $R$=0.3, in three coarse Pythia-compatible $p_T$ bins, for four iso cuts. For JETPHOX, coarse-bin values are cross-section weighted averages of the fine bins in Table 2.

| Generator | $p_T$ [GeV] | iso<1 | iso<2 | iso<3 | iso<4 |
|---|---|---|---|---|---|
| JETPHOX NLO | $[8, 14)$ | 0.8815 | 0.8180 | 0.7761 | 0.7471 |
| | $[14, 22)$ | 0.9203 | 0.8702 | 0.8322 | 0.7994 |
| | $[22, 36)$ | 0.9440 | 0.8956 | 0.8719 | 0.8354 |
| Pythia truth | $[8, 14)$ | 0.7611 | 0.7530 | 0.7485 | 0.7456 |
| | $[14, 22)$ | 0.8091 | 0.8011 | 0.7964 | 0.7933 |
| | $[22, 36)$ | 0.8312 | 0.8250 | 0.8211 | 0.8181 |
| JETPHOX − Pythia | $[8, 14)$ | +0.120 | +0.065 | +0.028 | +0.001 |
| | $[14, 22)$ | +0.111 | +0.069 | +0.036 | +0.006 |
| | $[22, 36)$ | +0.113 | +0.071 | +0.051 | +0.017 |

**Interpretation.** The headline finding is that JETPHOX and Pythia respond very differently to the iso cut:

- **JETPHOX $f_{\text{direct}}$ is strongly iso-dependent.** Across all three coarse $p_T$ bins, moving from iso<4 to iso<1 GeV raises $f_{\text{direct}}$ by $\sim 12-13$ percentage points. This is the expected behaviour of a fixed-order NLO calculation where the fragmentation contribution has an unconstrained $z \to 1$ endpoint and tends to have more hadronic activity in the cone, so tightening the iso cut preferentially suppresses it.

- **Pythia $f_{\text{direct}}$ is weakly iso-dependent.** The same iso tightening shifts Pythia's HepMC-classified $f_{\text{direct}}$ by only $\sim 1$ percentage point. Pythia's direct-vs-frag separation is at the parton-shower-history level rather than at the kinematic level: whether a class-2 (fragmentation) photon sits at iso=0.5 or iso=3 depends on the UE/MPI activity in the event, not on whether the photon was produced by a $2 \to 2$ matrix element.

- **The JETPHOX−Pythia offset closes as the cut relaxes.** At iso<4 the two agree to within $+0.001-0.017$; at iso<1 JETPHOX is $\sim +0.11-0.12$ higher. Equivalently: JETPHOX's "direct" definition becomes progressively more stringent as the iso cut tightens

(it is a kinematic+process-level definition), while Pythia's "direct" definition is fixed by the shower history (process-level only). The two definitions are not the same observable and should not be expected to agree away from a loose iso cut.

**Consequence for the measurement.** The Pythia truth fraction is what matters for the PPG12 efficiency denominator, and its weak iso-dependence means the denominator is robust against variations of the truth iso threshold. JETPHOX is the appropriate theoretical comparison for the reported cross section (which is $\sigma_{\text{direct}}+\sigma_{\text{frag}}$ under iso<4 GeV); the iso-cut scan here provides a useful check that the JETPHOX total cross section over 8−36 GeV varies by only a few percent over iso$\in [1, 4]$ GeV, consistent with the small iso tail of the NLO prediction. No re-weighting between Pythia and JETPHOX is implied.

## 5 Impact on the PPG12 measurement

**Efficiency denominator is consistent.** `RecoEffCalculator.C` applies the filter `particle_photonclass[ip` `< 3` at line 1252, which keeps both direct (class 1) and fragmentation (class 2) photons and rejects decay photons. The efficiency denominator and therefore the reported fiducial cross section is the *sum* of isolated direct and fragmentation photons, not direct-only. This is the standard convention used by ATLAS and CMS (see `wiki/physics/techniques/isolation-methods.md`), and it is what theory predictions such as JETPHOX provide natively, so no action is needed.

**Fiducial robustness.** Because $f_{\text{direct}}$ is weakly iso-dependent (Table 1), a systematic variation of the truth iso threshold within a physically motivated window (e.g. 3−5 GeV) does not strongly change the mix of contributions. This is reassuring for the fiducial definition choice at 4 GeV: the measurement is not sitting on a sharp transition.

**Caveats.** The photon-gun samples are generated with a hard-photon filter at generation level; this enriches direct relative to an unbiased minimum-bias Pythia sample. The absolute value of $f_{\text{direct}}$ reported here is therefore a property of the PPG12 signal MC denominator and not a prediction for the inclusive, unbiased prompt-photon composition in minimum-bias *pp* collisions. For the denominator and for interpreting the PPG12 cross section, this is the correct quantity; for an absolute theoretical cross-section decomposition, a dedicated JETPHOX direct-vs-frag study is required (wiki/physics/theory/fragmentation-functions.md).

## 6 Artifacts

All outputs are under `reports/direct_vs_frag_truth_iso/`.

- Extraction script (Pythia): `scripts/extract_direct_vs_frag.py`
- Plotting script (Pythia): `scripts/plot_direct_vs_frag.py`
- Independent verification script (Pythia): `scripts/verify_fractions.py`
- Extraction script (JETPHOX): `scripts/extract_jetphox_direct_vs_frag.py`
- Plotting script (JETPHOX vs Pythia): `scripts/plot_jetphox_vs_pythia.py`
- Independent verification (JETPHOX): `scripts/verify_jetphox_fdirect.py`
- Histograms and scan tree (Pythia): `direct_vs_frag_truth.root`
- Histograms (JETPHOX): `jetphox_direct_vs_frag.root`

- Numerical tables: `fraction_scan.csv`, `jetphox_fraction.csv`
- Figures: `figures/iso_ET_truth_stacked.pdf`, `figures/f_direct_vs_iso_cut.pdf`, `figures/jetphox_`

No nominal analysis code was modified.

## 7 Reviewer pass matrix

| Artifact | 2a Physics | 2b Cosmetics | 2c Re-derivation |
|---|---|---|---|
| `extract_direct_vs_frag.py` | PASS | N/A | PASS |
| `extract_jetphox_direct_vs_frag.py` (4 iso) | PASS | N/A | PASS |
| `plot_jetphox_vs_pythia.py` (3-panel) | PASS | N/A | N/A |
| `iso_ET_truth_stacked.pdf` | N/A | PASS | N/A |
| `f_direct_vs_iso_cut.pdf` | N/A | PASS | N/A |
| `jetphox_vs_pythia.pdf` | N/A | PASS | N/A |
| $f_{\text{direct}}$ at iso<4 GeV (Pythia, 3 pT bins) | N/A | N/A | PASS (exact match) |
| $f_{\text{direct}}$ at iso$\in \{1,2,3,4\}$ GeV (JETPHOX, 48 bins) | N/A | N/A | PASS (48/48 bins, 0.000% |

No fix was required; `3.5 fix-validator` is not applicable.